

Taiyang Xu

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Current position	Université catholique de Louvain , Louvain-la-Neuve, Belgium Institut de Recherche en Mathématique et Physique Postdoctoral Researcher (Mentor: Prof. Christophe Charlier)	08/2026 – now
Past positions	Fudan University , Shanghai, China Department of Mathematics Postdoctoral Fellow (Mentor: Prof. Lun Zhang)	07/2024 – 06/2026
Education	Fudan University , Shanghai, China Ph.D. in Mathematics, supervisor Prof. Engui Fan Thesis title: “ <i>On the long-time asymptotics of the local and nonlocal mKdV equation under the nonzero background</i> ” China University of Mining and Technology , Xuzhou, China B.Sc. in Mathematics, Distinguished Honor Thesis title: “ <i>Inverse scattering theory and integrability on several kinds of nonlinear evolution equations</i> ”	09/2019 – 06/2024 09/2015 – 06/2019
Research interests	Integrable PDEs, Random matrices theory, Determinantal point processes, Orthogonal polynomials, Asymptotic analysis, Riemann-Hilbert (RH) problems, Special functions, Painlevé equations.	
Publications	Preprints - Moment generating function of the tacnode process <i>arXiv:2606.17771</i> - Critical asymptotics of the Camassa-Holm equation in the small dispersion limit at the point of gradient catastrophe (with Yiling Yang and Lun Zhang) <i>available upon request</i> - Confluent hypergeometric kernel determinant on multiple large intervals (with Lun Zhang and Zhengyang Zhao) <i>arXiv:2508.10463</i> Publications in refereed journals - On the large-time asymptotics of the defocusing mKdV equation with step-like initial data (with Yidan Zhang) <i>Mathematical Physics, Analysis and Geometry</i>, 29 (2026), 38. - Painlevé transcendents in the defocusing mKdV equation with non-zero boundary conditions (with Engui Fan and Zhaoyu Wang) <i>Communications in Mathematical Physics</i>, 406 (2025), 181. - Soliton resolution and asymptotic stability of N-soliton solutions for the defocusing mKdV equation with a non-vanishing background (with Engui Fan and Zechuan Zhang) <i>Physica D: Nonlinear Phenomena</i>, 472 (2025), 134526. - Transient asymptotics of the modified Camassa-Holm equation (with Yiling Yang and Lun Zhang) <i>Journal of the London Mathematical Society</i>, 110 (2024), e12967. - On the Cauchy problem of defocusing mKdV equation with finite density initial data: long-time asymptotics in soliton-less regions (with Engui Fan and Zechuan Zhang) <i>Journal of Differential Equations</i>, 372 (2023), 55–122.	

6. Large-time asymptotics to the focusing nonlocal modified Kortweg-de Vries equation with step-like boundary conditions (with Engui Fan)
Studies in Applied Mathematics, 150 (2023), 1217–1273.
7. Riemann-Hilbert approach for multisoliton solutions of generalized coupled fourth-order nonlinear Schrödinger equations (with Weiqi Peng and Shoufu Tian)
Mathematical Methods in the Applied Sciences, 43 (2020), 865–880.

Grants

- Shanghai Post-doctoral Excellence Program (Grant No. 2024100) 07/2024 – 06/2026
Project: Riemann-Hilbert method for several asymptotic problems related to universality from integrable systems and random matrix theory
Role: Principal Investigator
- China Postdoctoral Science Foundation (Grant No. 2024M760480) 07/2024 – 06/2026
Project: Semiclassical asymptotics and universality for nonlinear integrable shallow water wave systems
Role: Principal Investigator

Teaching

09/2019 – 07/2026 @Fudan

- Fall, 2025: Recitation & Tutorial Instructor of Mathematical Analysis III (MATH130001.04).
- Spring, 2024: TA of Methods of Asymptotic Analysis (MATH630117).
- Fall, 2021: TA of Calculus A (MATH120021.02).
- Spring, 2020: TA of Calculus B (MATH120004.01) (Online).
- Fall, 2019: TA of Calculus B (MATH120003.01).

Research talks

- “On the Fredholm determinants” (invited), Seminar @Jiangsu Normal University, Xuzhou, China, 23 Jun, 2026.
- “Dubrovin’s conjecture for the Camassa-Holm equation” (invited), 3rd Workshop on Integrable Systems and Random Matrix Theory 2026, Dongguan, China, 4–17 Jan, 2026.
- “Confluent hypergeometric kernel determinant on multiple large intervals” (invited), Bielefeld-Melbourne-Seoul Online Random Matrix Seminar, 15th Oct, 2025.
- “Confluent hypergeometric kernel determinant on multiple large intervals” (contributed), 5th ZiF Summer School “Randomness in Physics and Mathematics: From Thermalisation in Quantum Systems to Random Matrices”, Universität Bielefeld, Bielefeld, Germany, 25 Aug–6 Sep, 2025.
- “An introduction to Dubrovin’s universality conjecture for integrable PDEs, and some progress on the Camassa-Holm equation” (invited), Seminar @UCLouvain, Louvain-la-Neuve, Belgium, 21 Aug, 2025.
- “Transient asymptotics of the modified Camassa-Holm equation” (invited), The 2nd Workshop on Integrable Systems and Random Matrix Theory, Dongguan, China, 5–17 Jan, 2025.
- “Integrable PDEs with nonzero boundary conditions: large-time asymptotics” (contributed), The 15th Hemudu Forum on Integrable Systems, Ningbo, China, 24–26 Nov, 2023.

Other presentations Outreach talks

- “Some asymptotic problems in mathematical physics” (invited), Shanghai Institute of Technical Physics, Shanghai, China, 29th Apr, 2025.

12 **presentations** (including reading reports) given at Fudan Integrable Systems and Random Matrix Theory Seminar.

Academic visits	– 22/06/2026 – 25/06/2026, Jiangsu Normal University, China. (Host: Dong Yao) – 18/08/2025 – 24/08/2025, UCLouvain, Belgium. (Host: Christophe Charlier) – 31/03/2025 – 11/04/2025, Chongqing University, China. (Host: Yiling Yang) (I greatly appreciate for their warm hospitality)
Service	Co-organizer of the following activities – (with Lun Zhang) Weekly RMT seminar @Fudan, 09/2022 – 06/2026. – (with Junwen Liu and Lun Zhang) Mini-workshop on Random Matrices, Fudan University, Shanghai, China, 4 June, 2026. – (with Engui Fan, Zhaoyu Wang, Yiling Yang and Lun Zhang) Workshop on Integrable Systems, Fudan University, Shanghai, China, 28 Nov, 2025. – (with Lun Zhang) Mini-workshop on Asymptotic Analysis, Fudan University, Shanghai, China, 5–9 Jun, 2025. Referee services provided for the following journals – Analysis and Applications – Journal of Nonlinear Mathematical Physics – Qualitative Theory of Dynamical Systems – Studies in Applied Mathematics
Scholarships and awards	09/2019 – 06/2024 @Fudan (Doctorate) – Shanghai Outstanding Graduate (Graduation with Honors), 2024. – Huatai Securities Science and Technology Scholarship, 2023. – Pacific Insurance Company Scholarship, 2022. – Outstanding Doctoral Candidate Scholarship, Fudan University, 2021. – Fudan University Doctoral Scholarship, 2019–2023. 09/2015 – 06/2019 @CUMT (Undergraduate) – Outstanding Undergraduates in China University of Mining and Technology, 2019.
Status	China – citizen, Married .
Languages	– Chinese (native), The Qianwei dialect of Sichuanese (native) – English
Computer skills	$\LaTeX$, Mathematica, Matlab, HTML, C++, Javascript

Thursday 6th August, 2026